Global CO_2 Emissions:

Trends, Regional Shifts, and Per Capita Inequalities

Scope: 1750–2024

A data-driven brief for policy analysts and climate negotiators

Sources: Global Carbon Budget 2025 (Our World in Data), NASA/NOAA

Atmospheric CO_2 Reaches 429 ppm — 50% Above Pre-Industrial Levels

429 ppm

NOAA Mauna Loa, Feb 2026

- Pre-industrial baseline: ~280 ppm
- Increase: more than 50% in roughly two centuries
- Current level is at least 1.5× greater than natural end-of-ice-age rise over millennia
- Primary driver: fossil fuel combustion

1750: ~280 ppm

1958: Keeling record begins

2026: 429 ppm

Sources: Global Carbon Budget 2025 (Our World in Data), NASA/NOAA

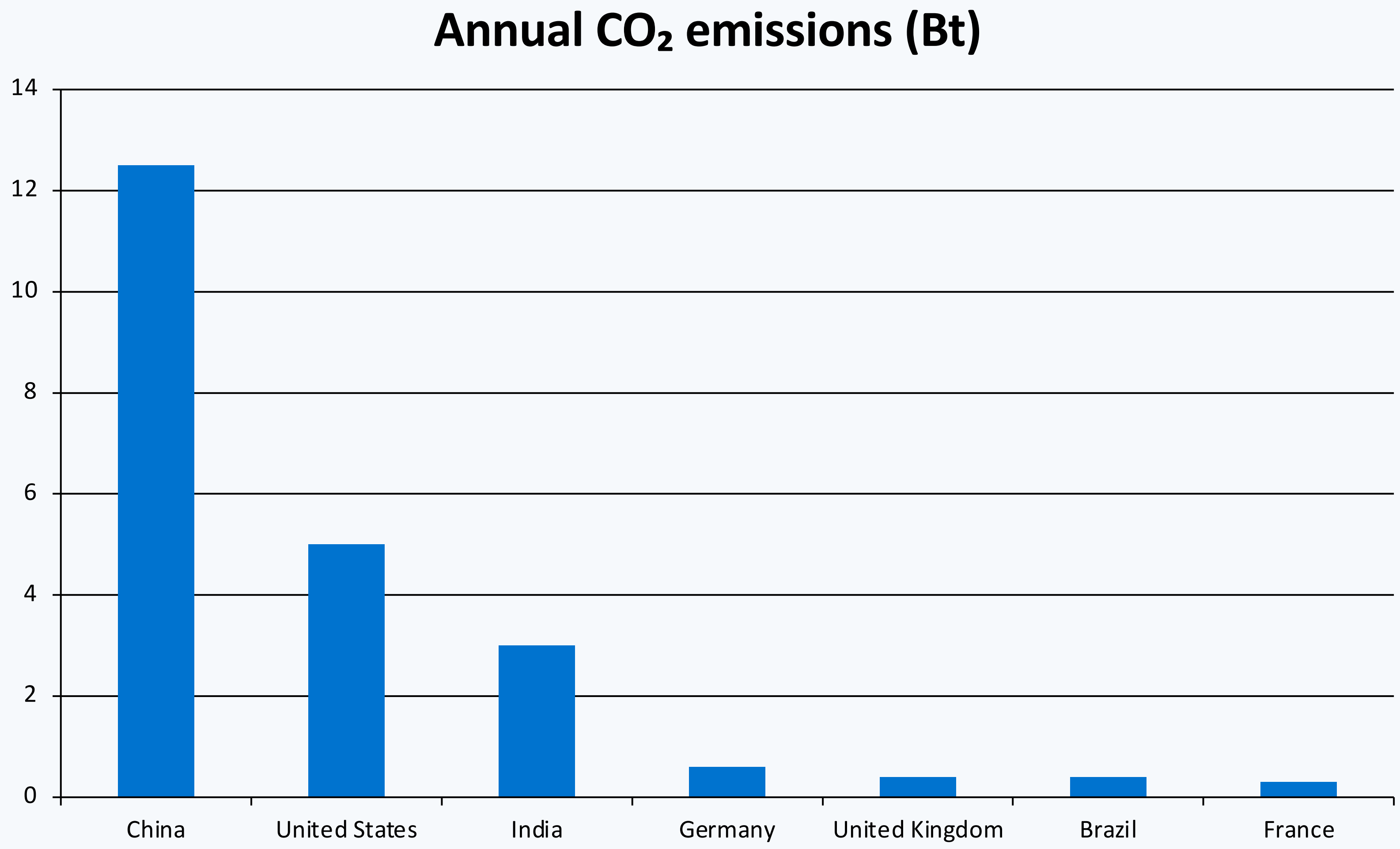
Global Fossil Fuel CO_2 Emissions Grew Six-Fold: 6 Billion to 35+ Billion Tonnes

Inflection points

- ~6 Bt in 1950
- ~20 Bt by 1990
- >35 Bt by 2024
- Growth slowed but has not peaked

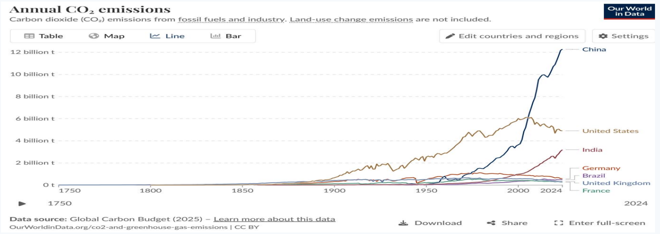
Sources: Global Carbon Budget 2025 (Our World in Data), NASA/NOAA

China Now Emits Over 12 Billion Tonnes — More Than Double the United State



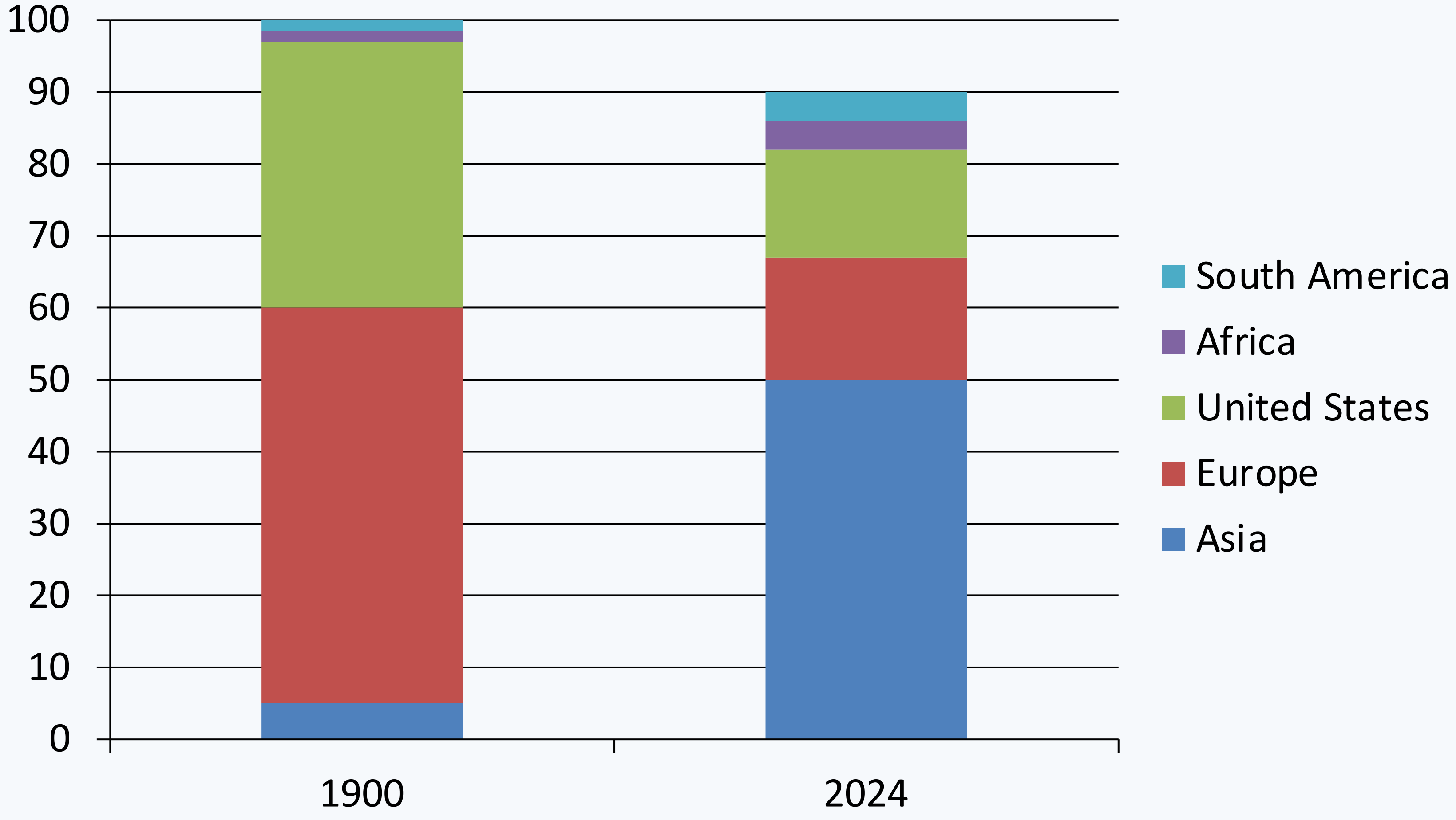
2024 snapshot

- China: ~12.5 Bt
- US: ~5 Bt
- India: ~3 Bt
- China now >2× US
- US declined from ~6 Bt peak



Sources: Global Carbon Budget 2025 (Our World in Data), NASA/NOAA

Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Thir

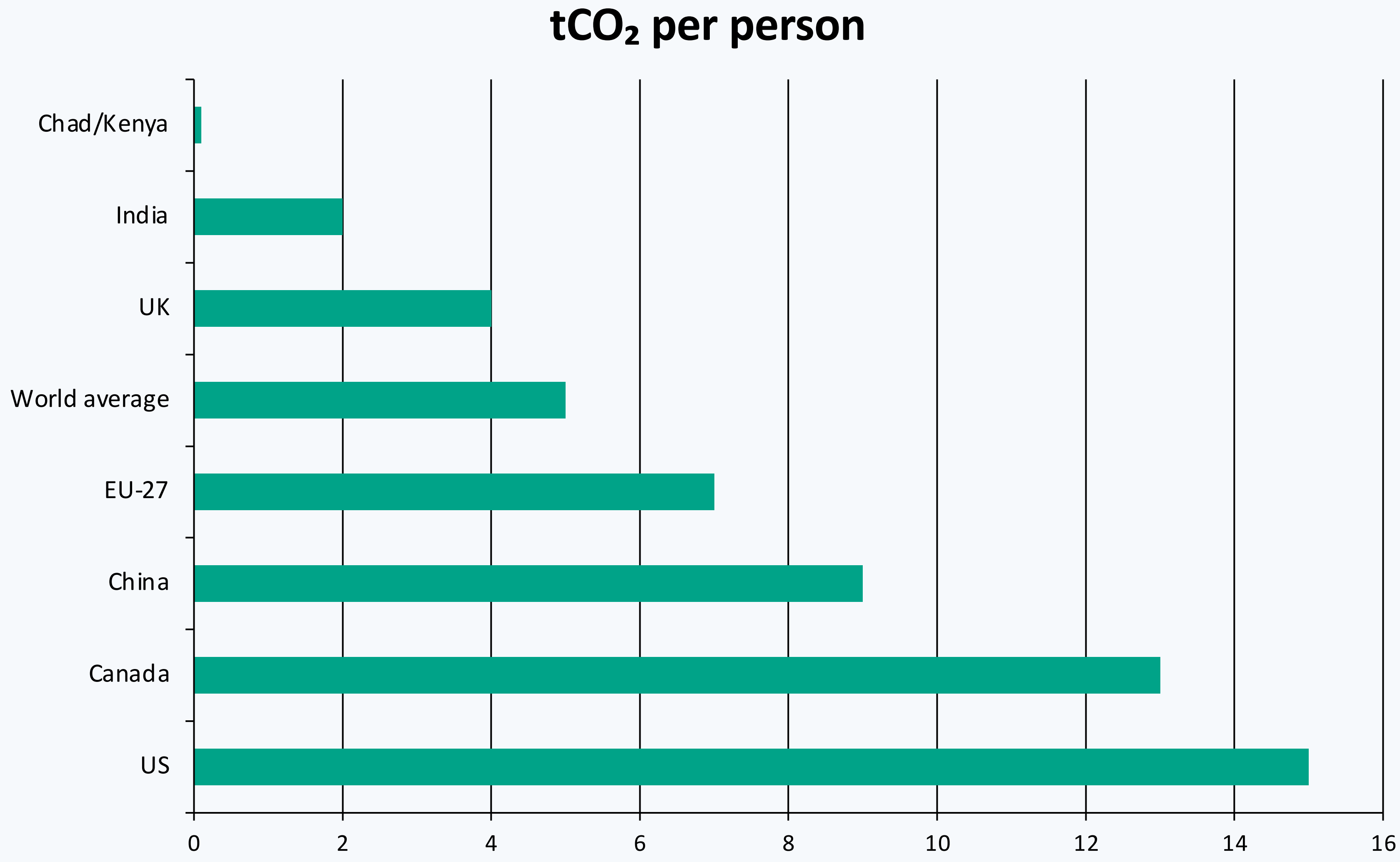


Shift in shares

- 1900: Europe + US >90%
- 2024: Asia ~50%
- US + Europe <33%
- Africa & South America each ~3–4%

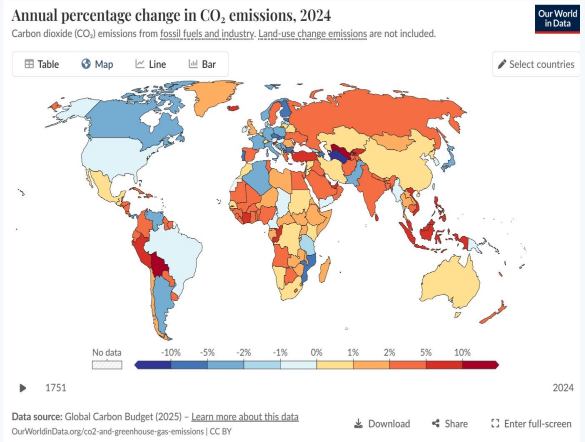
Sources: Global Carbon Budget 2025 (Our World in Data), NASA/NOAA

Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity



≈150×

US vs Chad



Sources: Global Carbon Budget 2025 (Our World in Data), NASA/NOAA

2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase

Map interpretation

- US & much of Europe: declines (roughly −1% to −5%, blue)
- Russia, parts of Southeast Asia, and several African countries: increases (+2% to +10%+, orange/red)
- Mixed patterns in South America

Sources: Global Carbon Budget 2025 (Our World in Data), NASA/NOAA

Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Countries with similar prosperity can diverge sharply in per capita CO_2.

Differences reflect power-sector choices: nuclear, renewables,
or continued dependence on fossil fuels.
This is a policy-sensitive, not purely income-driven, outcome.

Country	Energy mix profile	Per capita CO_2
France	High nuclear + growing renewables	**~4 t**
United Kingdom	High renewables, coal phase-out	**~4 t**
Germany	Higher fossil share despite transition	**~7 t**
Netherlands	Higher gas/fossil share	**~7 t**

Sources: Global Carbon Budget 2025 (Our World in Data), NASA/NOAA

Key Takeaways: Emissions Are Still Rising, But the Map of Responsibility Is Shift

- Global CO_2 emissions exceed 35 billion tonnes per year and have not yet peaked.
- China emits over one-quarter of the global total (~12.5 Bt), more than double the US (~5 Bt).
- Regional composition has transformed: Asia now contributes ~50%, while US + Europe are below one-third.
- Per capita inequality remains extreme: ~15 t/person in the US versus ~0.1 t/person in Chad (≈150×).
- Energy mix and policy choices materially shape outcomes among countries with similar living standards.

Implication: negotiations must balance total emissions, per capita equity, and transition pathways simultaneously.

Sources: Global Carbon Budget 2025 (Our World in Data), NASA/NOAA